

Whole-Person Health States and Outcomes

Symptoms, function, quality of life, resilience, multimorbidity, goals, trajectories, and integrated profiling

Volume 01 of the Health Atlas Research Suite

Purpose

Define the outcomes the entire suite is meant to explain and improve, preventing organ metrics or diagnostic codes from replacing lived health.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume is a framework for integrated assessment, not a universal wellness score, diagnostic shortcut or individualized treatment plan. Urgent symptoms still require organ- and syndrome-specific evaluation.

How to use this volume

A whole-person profile is a longitudinal model of biological state, symptoms, function, capability, participation, well-being, risk, treatment burden, context and goals. It explicitly allows discordance: a person may have normal routine tests and disabling symptoms, severe pathology and preserved function, or multiple “controlled” diseases with intolerable care burden. The profile should support action without reducing a person to one score.

Questions this volume is designed to answer

1. What outcomes should health systems and devices ultimately improve?
2. How can symptoms, physiology and function disagree?
3. How should multimorbidity and treatment burden be represented?
4. What measurements preserve person-level meaning and uncertainty?
5. How should goals, context and rights enter technical profiles?

Scope

- Health states, symptoms, function, participation, quality of life, reserve, resilience and vulnerability.
- Acute/chronic illness, multimorbidity, persistent symptoms, pain, disability, frailty, recovery, serious illness and flourishing.
- PROMs, performance, wearables, contextual and integrated longitudinal measurement.
- Goal-oriented, rehabilitative, preventive, social, palliative and system interventions.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

Contents

Document status and intended use	i
How to use this volume	ii
1 Executive synthesis	1
1.1 What a complete profile requires	1
2 Historical development and changing models	3
2.0.1 Health was interpreted through balance, virtue, community, spirituality and the person's place in the world.	3
2.0.2 Anatomy and lesion-based pathology organized medicine around organs and diseases.	4
2.0.3 Vital statistics, laboratory tests, randomized trials and risk factors quantified disease and outcomes.	4
2.0.4 Biopsychosocial models, disability frameworks, quality-of-life science and evidence-based medicine restored multiple dimensions.	4
2.0.5 Digital phenotyping, multimodal records, precision medicine and causal methods enable longitudinal integration.	4
2.0.6 Whole-person health is represented as a multi-axial, longitudinal, goal-sensitive state across biology, function and context.	4
3 Normal foundations: structure, physiology, and regulation	5
3.0.1 Health as a multidimensional state	5
3.0.2 Symptoms and interoception	5
3.0.3 Function, activity and participation	6
3.0.4 Quality of life and well-being	6
3.0.5 Reserve, resilience and vulnerability	6
3.0.6 Multimorbidity and treatment interaction	7
3.0.7 Trajectory and time	7
3.0.8 Goals, preferences and decision context	7
3.0.9 Social and environmental embodiment	8
3.0.10 Care burden and implementation	8
3.0.11 Uncertainty and diagnostic humility	8
3.1 Useful quantitative relationships	9
4 Determinants, heterogeneity, and causal architecture	10
4.0.1 Genetic architecture and family history	12
4.0.2 Development and life course	12
4.0.3 Nutrition and metabolism	12

4.0.4	Physical activity, conditioning, and sedentary behavior	13
4.0.5	Sleep, circadian timing, and recovery	13
4.0.6	Stress, emotion, and behavior	13
4.0.7	Social, economic, cultural, and care conditions	14
4.0.8	Environmental and occupational exposures	14
4.0.9	Health care, medication, and iatrogenic effects	14
5	Major disorders, failure modes, and clinically important phenotypes	16
5.0.1	Asymptomatic health with latent risk	16
5.0.2	Acute illness episode	16
5.0.3	Chronic disease with stable function	17
5.0.4	Multimorbidity and polypharmacy	17
5.0.5	Persistent physical symptoms with incomplete explanation	18
5.0.6	Chronic pain state	18
5.0.7	Disability with stable or progressive impairment	19
5.0.8	Frailty and high vulnerability	19
5.0.9	Recovery and rehabilitation trajectory	20
5.0.10	Relapsing or episodic condition	20
5.0.11	Treatment-related burden or iatrogenic illness	21
5.0.12	Serious illness and palliative needs	21
5.0.13	Flourishing and positive health	22
6	Assessment, clinical reasoning, and longitudinal profiling	23
6.0.1	1. Question and urgency	24
6.0.2	2. History and lived phenotype	24
6.0.3	3. Examination and basic measurements	24
6.0.4	4. Targeted laboratory, imaging, and physiology	25
6.0.5	5. Function and challenge testing	25
6.0.6	6. Longitudinal integration	25
7	Measurement and device landscape	26
7.0.1	Narrative history and longitudinal timeline	28
7.0.2	Physical and mental-status examination	28
7.0.3	Patient-reported outcome measures	28
7.0.4	WHODAS/ICF-based disability assessment	29
7.0.5	Performance-based capacity tests	29
7.0.6	Ecological momentary assessment	30
7.0.7	Wearable and ambient monitoring	30
7.0.8	Laboratory and imaging summary profile	30
7.0.9	Medication and intervention burden inventory	31

7.0.10 Social and environmental context assessment	31
7.0.11 Goal-attainment and preference elicitation	31
7.0.12 Composite risk and frailty models	32
7.0.13 Integrated health record / personal health graph	32
8 Engineering architectures, signal quality, and validation	33
8.0.1 Intended use and measurand	34
8.0.2 Sensor and transduction	34
8.0.3 Timing and synchronization	34
8.0.4 Analog, acquisition, and digital integrity	35
8.0.5 Calibration and traceability	35
8.0.6 Artifact and quality channels	35
8.0.7 Algorithms and uncertainty	36
8.0.8 Safety, usability, cybersecurity, and privacy	36
9 Prevention, treatment, rehabilitation, and system intervention	37
9.0.1 Upstream prevention and healthy environments	38
9.0.2 Shared decision and goal-oriented care	39
9.0.3 Integrated multimorbidity management	39
9.0.4 Rehabilitation and adaptive support	39
9.0.5 Symptom and pain management	39
9.0.6 Mental-health and behavioral intervention	40
9.0.7 Social and material support	40
9.0.8 Medication review and deprescribing	40
9.0.9 Caregiver and relationship support	40
9.0.10 Palliative and serious-illness care	41
9.0.11 Learning health-system follow-up	41
10 What is known, not known, and frequently overstated	42
10.0.1 Can whole-person health be summarized without losing meaning?	43
10.0.2 How should response shift be interpreted?	43
10.0.3 How much risk is genetic versus environmental?	44
10.0.4 Can one composite score summarize health?	44
10.0.5 When is earlier detection beneficial?	44
10.0.6 Do digital biomarkers generalize?	45
10.0.7 What counts as meaningful change?	45
11 Research and development roadmap	46
11.0.1 0. Whole-person common outcome set	47
11.0.2 0. Goal-concordant multimorbidity trial	47
11.0.3 1. Reference map	47

11.0.4 2. Causal determinants	48
11.0.5 3. Measurement validity	48
11.0.6 4. Clinical utility	48
11.0.7 5. Longitudinal multimorbidity	49
11.0.8 6. Implementation and governance	49
12 Cross-volume integration	50
A Minimum data schema for a longitudinal domain profile	51
B Glossary	52
Selected references and current standards	53

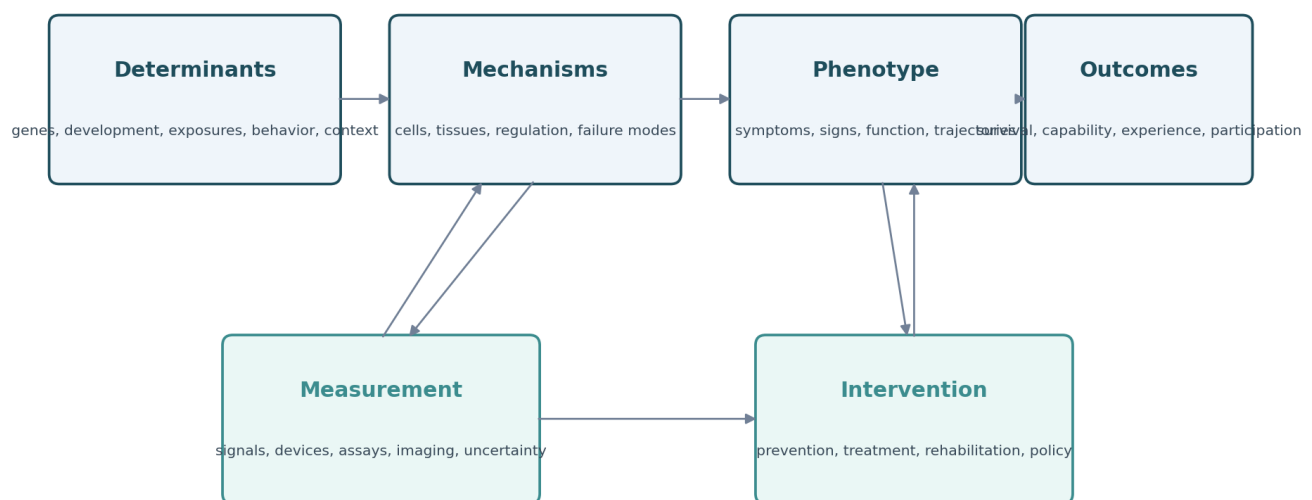
Executive synthesis

A whole-person profile is a longitudinal model of biological state, symptoms, function, capability, participation, well-being, risk, treatment burden, context and goals. It explicitly allows discordance: a person may have normal routine tests and disabling symptoms, severe pathology and preserved function, or multiple “controlled” diseases with intolerable care burden. The profile should support action without reducing a person to one score.

Central organizing principle

Define the outcomes the entire suite is meant to explain and improve, preventing organ metrics or diagnostic codes from replacing lived health.

Whole-Person Health States and Outcomes



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.

- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Whole-person framing must not delay urgent evaluation of dangerous symptoms or be used to dismiss disease as “psychosocial.” Conversely, abnormal tests without symptoms or benefit should not automatically trigger invasive cascades.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early holistic and moral models	Health was interpreted through balance, virtue, community, spirituality and the person's place in the world.	Preserved lived meaning and context.	Biological mechanisms and measurement were limited; moral blame was common.
Anatomical and pathological organization	Anatomy and lesion-based pathology organized medicine around organs and diseases.	Enabled precise diagnosis and treatment.	Function, suffering and social causes were displaced.
Statistical and laboratory medicine	Vital statistics, laboratory tests, randomized trials and risk factors quantified disease and outcomes.	Made prevention and evidence comparison possible.	What was easy to count often became what mattered.
Biopsychosocial and evidence-based expansion	Biopsychosocial models, disability frameworks, quality-of-life science and evidence-based medicine restored multiple dimensions.	Created validated patient-reported and functional outcomes.	Practice and reimbursement remained disease-centered.
Systems, data and precision era	Digital phenotyping, multimodal records, precision medicine and causal methods enable longitudinal integration.	Can connect mechanisms to daily life.	Data volume, bias and composite-score overreach threaten meaning.
Current integrated model	Whole-person health is represented as a multi-axial, longitudinal, goal-sensitive state across biology, function and context.	Provides the outcome layer for every suite volume.	Institutions still optimize specialty metrics rather than total benefit.

2.0.1 Health was interpreted through balance, virtue, community, spirituality and the person's place in the world.

Scientific consequence. Preserved lived meaning and context.

What remained unresolved. Biological mechanisms and measurement were limited; moral blame was common.

Evidence anchors: ^[1,3]

2.0.2 Anatomy and lesion-based pathology organized medicine around organs and diseases.

Scientific consequence. Enabled precise diagnosis and treatment.

What remained unresolved. Function, suffering and social causes were displaced.

Evidence anchors: ^[1,3]

2.0.3 Vital statistics, laboratory tests, randomized trials and risk factors quantified disease and outcomes.

Scientific consequence. Made prevention and evidence comparison possible.

What remained unresolved. What was easy to count often became what mattered.

Evidence anchors: ^[1,3]

2.0.4 Biopsychosocial models, disability frameworks, quality-of-life science and evidence-based medicine restored multiple dimensions.

Scientific consequence. Created validated patient-reported and functional outcomes.

What remained unresolved. Practice and reimbursement remained disease-centered.

Evidence anchors: ^[1,3]

2.0.5 Digital phenotyping, multimodal records, precision medicine and causal methods enable longitudinal integration.

Scientific consequence. Can connect mechanisms to daily life.

What remained unresolved. Data volume, bias and composite-score overreach threaten meaning.

Evidence anchors: ^[1,3]

2.0.6 Whole-person health is represented as a multi-axial, longitudinal, goal-sensitive state across biology, function and context.

Scientific consequence. Provides the outcome layer for every suite volume.

What remained unresolved. Institutions still optimize specialty metrics rather than total benefit.

Evidence anchors: ^[1,5]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Health as a multidimensional state

Core principle. Health includes survival, symptoms, physiology, function, capability, participation, quality of life, meaning and context rather than a binary absence of diagnosis.

Control architecture. Biology, development, behavior, relationships, environment, resources and care interact over time.

Failure modes. A person can have disease without disability, disability without a single disease label, or favorable biomarkers with severe lived burden.

Observable consequences. Mortality/morbidity, symptoms, function, PROMs, goals, participation, resource use and context.

Cross-system coupling. Every atlas branch.

Evidence anchors: [1,3,5]

3.0.2 Symptoms and interoception

Core principle. Symptoms emerge when bodily signals are generated, transmitted, interpreted and assigned salience within cognitive and social context.

Control architecture. Peripheral physiology, neural prediction, attention, memory, language, culture and prior learning.

Failure modes. Under-recognition, amplification, dissociation, persistent symptoms or discordance with routine tests.

Observable consequences. Structured history, symptom dimensions, organ testing as indicated, repeated context and functional impact.

Cross-system coupling. Nervous, cognitive, psychosocial, pain and organ systems.

Evidence anchors: [4,28,29]

3.0.3 Function, activity and participation

Core principle. Body capacity becomes real-world activity and social participation only through tasks, support and environment.

Control architecture. Organ reserve, skills, assistive products, built environment, roles, policy and adaptation.

Failure modes. Impairment, activity limitation, participation restriction or environmental disability.

Observable consequences. ICF/WHODAS, performance tests, activity, role participation and goals.

Cross-system coupling. Rehabilitation, disability rights, social systems and all organs.

Evidence anchors: [5,6,7]

3.0.4 Quality of life and well-being

Core principle. People value comfort, autonomy, relationships, purpose, security and opportunity in addition to disease outcomes.

Control architecture. Symptoms, function, expectations, adaptation, resources, identity and social comparison.

Failure modes. Response shift, distress, loss of meaning, treatment burden or outcome measures that miss what matters.

Observable consequences. WHOQOL, PROMIS, preference-sensitive and individualized goal measures.

Cross-system coupling. Psychosocial, mental health, palliative and ethics domains.

Evidence anchors: [3,4,25]

3.0.5 Reserve, resilience and vulnerability

Core principle. The capacity to tolerate challenge and recover depends on organ reserve, cognition, mobility, nutrition, social support and environment.

Control architecture. Redundancy, plasticity, conditioning, allostasis, resources and prior exposures.

Failure modes. Frailty, decompensation, slow recovery, relapse or disproportionate harm from small stressors.

Observable consequences. Challenge tests, frailty/function, recovery slope, variability and support network.

Cross-system coupling. Homeostasis, aging, rehabilitation and social connection.

Evidence anchors: [9,12,13]

3.0.6 Multimorbidity and treatment interaction

Core principle. Multiple conditions, medications and care pathways interact nonadditively within one person.

Control architecture. Shared causes, organ crosstalk, competing risks, polypharmacy and conflicting recommendations.

Failure modes. Single-disease optimization can worsen total burden, function or goals.

Observable consequences. Condition/treatment timeline, interactions, burden, function, priorities and care coordination.

Cross-system coupling. Health systems, therapeutics, aging and measurement.

Evidence anchors: [24,32]

3.0.7 Trajectory and time

Core principle. Health is a path through baseline, risk, onset, compensation, crisis, recovery, adaptation, recurrence and decline.

Control architecture. Development, cumulative exposure, interventions, feedback and stochastic events.

Failure modes. Snapshot misclassification, survivor bias, delayed diagnosis or false recovery.

Observable consequences. Absolute time, life stage, episodes, event markers, repeated measurements and change uncertainty.

Cross-system coupling. Life course, data science and all clinical domains.

Evidence anchors: [14,15,31]

3.0.8 Goals, preferences and decision context

Core principle. The best action depends on what outcomes the person values, acceptable burden, uncertainty and alternatives.

Control architecture. Values, prognosis, autonomy, family roles, culture, cost and prior experience.

Failure modes. Preference misdiagnosis, paternalism, decisional regret or burdensome low-value care.

Observable consequences. Goal elicitation, shared decision quality, advance care plans and outcome concordance.

Cross-system coupling. Ethics, palliative, delivery systems and therapeutics.

Evidence anchors: [25,37,38]

3.0.9 Social and environmental embodiment

Core principle. Housing, work, discrimination, relationships, food, air, violence and access become biological and functional exposures.

Control architecture. Material conditions, stress physiology, behavior opportunity, direct toxic/infectious exposure and care pathways.

Failure modes. Inequity, clustered risk, chronic stress and preventable disease/disability.

Observable consequences. Individual and area exposure, social needs/resources, structural measures and longitudinal outcomes.

Cross-system coupling. Psychosocial, environmental, public health and equity.

Evidence anchors: [19,20,21]

3.0.10 Care burden and implementation

Core principle. Appointments, monitoring, side effects, self-management, cost and administrative complexity are health exposures.

Control architecture. Workload-capacity mismatch, access, coordination and treatment opportunity cost.

Failure modes. Nonadherence mislabeled as individual failure, burnout, financial toxicity or fragmented care.

Observable consequences. Treatment workload, costs, time, access, adherence context and patient/caregiver experience.

Cross-system coupling. Health systems, therapeutics, ethics and social context.

Evidence anchors: [22,23,24]

3.0.11 Uncertainty and diagnostic humility

Core principle. Every health statement carries uncertainty from biology, measurement, evidence and values.

Control architecture. Pretest probability, test error, causal ambiguity, missingness, model drift and preference variation.

Failure modes. Overdiagnosis, underdiagnosis, false reassurance, unnecessary cascades or nihilism.

Observable consequences. Probability, intervals, quality flags, alternative explanations and follow-up plan.

Cross-system coupling. Measurement/evidence, ethics and all domains.

Evidence anchors: [16,17,18]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Quality-adjusted life concept	$QALY = \int_0^T u(t) dt$	Combines time with a preference weight for health state in economic evaluation.	Weights are value-laden, population-dependent and cannot replace individual goals or rights.
Reliable change index	$RCI = \frac{X_2 - X_1}{S_{diff}}$	Compares observed score change with expected measurement error.	Requires valid reliability and does not establish clinical importance.
Absolute risk reduction	$ARR = Risk_{control} - Risk_{treatment}$	Expresses outcome difference on an absolute scale relevant to decisions.	Depends on baseline risk, time horizon, competing risks and study transportability.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within whole-person health states and outcomes.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within whole-person health states and outcomes.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[14]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[11]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[11]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[13]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: [13,19]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: [19,21]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: [20]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[24,40]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the whole-person health state, trajectory, lived-experience phenotype, or outcome pattern, but cross-system processes and functional effects are retained.

5.0.1 Asymptomatic health with latent risk

Definition and phenotype. No current symptom or functional loss despite inherited, exposure, biomarker or imaging risk.

Pathophysiology. Risk factors may alter probability through causal pathways without guaranteeing disease.

Determinants and modifiers. Age, genetics, exposures, social context, screening history and competing risks.

Assessment. Validated risk estimation, modifiable determinants and screening only where net benefit is established.

Intervention logic. Proportionate prevention and follow-up rather than treating every abnormality as disease.

Known limits and open questions. Risk models can overstate certainty, produce labeling and change with population/time.

Suite links. 11 prevention; 13 evidence; organ volumes.

Evidence anchors: ^[11,16,17]

5.0.2 Acute illness episode

Definition and phenotype. Rapid change from baseline due to infection, injury, organ failure, toxin or other stressor.

Pathophysiology. Threat, physiologic compensation, care response and preexisting reserve shape outcome.

Determinants and modifiers. Cause, severity, age, comorbidity, access, delay and system quality.

Assessment. Urgency, causal syndrome, organ function, baseline and response trajectory.

Intervention logic. Treat cause and support function while preventing iatrogenic harm; plan recovery early.

Known limits and open questions. Severity scores summarize groups but may not identify individual reversibility or goals.

Suite links. 03O homeostasis; 12 acute care; 10 recovery.

Evidence anchors: [5,24]

5.0.3 Chronic disease with stable function

Definition and phenotype. Persistent pathophysiology managed sufficiently to preserve valued activity.

Pathophysiology. Compensation, treatment, behavior/environment and adaptation keep demand within reserve.

Determinants and modifiers. Disease type, self-management capacity, treatment burden, access and social support.

Assessment. Disease control plus symptoms, function, adverse effects and trajectory.

Intervention logic. Prevention of progression, low-burden management and periodic goal review.

Known limits and open questions. “Stable” may conceal declining reserve or high treatment workload.

Suite links. Organ volumes; 14 therapeutics; 01 whole-person.

Evidence anchors: [4,5,22]

5.0.4 Multimorbidity and polypharmacy

Definition and phenotype. Coexisting chronic conditions and treatments with interacting benefits, harms and priorities.

Pathophysiology. Shared mechanisms and competing interventions create nonlinear total effects.

Determinants and modifiers. Age, social disadvantage, fragmented specialties and survival with chronic disease.

Assessment. Medication/condition timeline, interactions, function, prognosis, burden and priorities.

Intervention logic. Integrated goal-oriented care, deprescribing where appropriate and coordinated monitoring.

Known limits and open questions. Trials and guidelines often exclude complex patients; optimal combinations are uncertain.

Suite links. 12 delivery; 14 therapeutics; 07 aging.

Evidence anchors: [16,24,32]

5.0.5 Persistent physical symptoms with incomplete explanation

Definition and phenotype. Pain, fatigue, dizziness, GI or other symptoms persist despite no single sufficient routine diagnosis.

Pathophysiology. Undetected disease, post-acute change, autonomic/immune/metabolic factors, sensitization, sleep, stress and learning can coexist.

Determinants and modifiers. Prior illness/injury, trauma, healthcare experience, social stress, sleep and comorbidity.

Assessment. Validate symptoms; rule out urgent/treatable disease; characterize triggers, function, recovery and cross-system phenotype.

Intervention logic. Treat identified mechanisms, rehabilitate/pacing as tolerated and integrate psychological/social support without false dichotomy.

Known limits and open questions. Absence of a routine marker is not proof of psychogenesis; broad labels mix mechanisms.

Suite links. 03O homeostasis; 05/06; 10 rehab.

Evidence anchors: [9,28,29]

5.0.6 Chronic pain state

Definition and phenotype. Pain persisting or recurring beyond expected healing, with nociceptive, neuropathic and/or nociplastic mechanisms.

Pathophysiology. Tissue/nerve signals interact with sensitization, prediction, sleep, mood, behavior and context.

Determinants and modifiers. Injury/disease, stress/trauma, work, treatment exposure, inequality and prior pain.

Assessment. Mechanism-informed assessment, red flags, function, sleep/mood and patient goals.

Intervention logic. Multimodal low-harm care emphasizing function, mechanism and shared goals; procedures/drugs only when benefit is plausible.

Known limits and open questions. Pain intensity does not reveal mechanism and biomarkers remain limited.

Suite links. 03I MSK; 05 brain; 06 psychosocial; 14 therapy.

Evidence anchors: [3,4,29]

5.0.7 Disability with stable or progressive impairment

Definition and phenotype. Long-term interaction between impairment and environmental/personal barriers affecting activity or participation.

Pathophysiology. Body difference combines with inaccessible design, stigma, policy, support and adaptation.

Determinants and modifiers. Congenital/acquired condition, resources, assistive technology, social attitudes and rights enforcement.

Assessment. Self-defined goals, ICF/WHODAS, task/environment assessment and health maintenance.

Intervention logic. Accessibility, assistive products, rehabilitation and rights-based services; avoid cure-only framing.

Known limits and open questions. Medical measures often underrepresent participation and positive disability identity.

Suite links. 10 function; 15 rights; 06 culture.

Evidence anchors: [5,7,8]

5.0.8 Frailty and high vulnerability

Definition and phenotype. Reduced reserve across multiple systems causing disproportionate risk from ordinary stress.

Pathophysiology. Cumulative deficits, muscle/cognition/nutrition/social factors and multimorbidity interact.

Determinants and modifiers. Age, inactivity, hospitalization, poverty/isolation, medications and disease.

Assessment. Frailty plus reversible causes, baseline function/cognition, support and goals.

Intervention logic. Strength/nutrition/mobility, medication and environment optimization, prevention and advance planning.

Known limits and open questions. Frailty is dynamic and scores should not deny beneficial care.

Suite links. 07 aging; 03O reserve; 10 rehab.

Evidence anchors: [5,9,12]

5.0.9 Recovery and rehabilitation trajectory

Definition and phenotype. Period after illness, injury or treatment in which function returns, adapts or remains altered.

Pathophysiology. Repair, plasticity, conditioning, symptom control, learning, support and environment determine slope.

Determinants and modifiers. Severity, baseline reserve, rehabilitation access, mental health, work/family demands and iatrogenesis.

Assessment. Baseline/serial function, symptoms, exertional response, goals, environment and participation.

Intervention logic. Early individualized rehabilitation, assistive/adaptive support and coordinated follow-up.

Known limits and open questions. Biomarker normalization may precede or fail to predict lived recovery.

Suite links. 10 rehabilitation; 12 transitions; organ volumes.

Evidence anchors: [4,5,9]

5.0.10 Relapsing or episodic condition

Definition and phenotype. Health alternates between attacks and partial/full remission, often with uncertain triggers.

Pathophysiology. Thresholded network dynamics, immune/neural/endocrine cycles, exposure and treatment adherence may generate episodes.

Determinants and modifiers. Season, infection, stress, behavior, drugs and underlying susceptibility.

Assessment. Event diary, objective markers during/ between episodes, triggers and cumulative burden.

Intervention logic. Prevent attacks, rapid action plan and reduce residual burden/uncertainty.

Known limits and open questions. Recall and post hoc trigger attribution are biased; between-episode tests may be normal.

Suite links. 03O dynamics; 13 longitudinal methods.

Evidence anchors: [15,30]

5.0.11 Treatment-related burden or iatrogenic illness

Definition and phenotype. Symptoms, injury or disability caused or worsened by medication, procedure, device or healthcare process.

Pathophysiology. Expected toxicity, interaction, error, infection, deconditioning, diagnostic cascade or system failure.

Determinants and modifiers. Polypharmacy, transitions, fragmented records, invasive intensity and vulnerable reserve.

Assessment. Temporal treatment map, adverse-event assessment, alternatives and system review.

Intervention logic. Stop/modify harm, supportive/rehabilitative care, disclosure and prevention redesign.

Known limits and open questions. Causality is difficult and harms are underreported.

Suite links. 12 patient safety; 14 pharmacology; 15 accountability.

Evidence anchors: ^[24]

5.0.12 Serious illness and palliative needs

Definition and phenotype. Life-limiting or high-burden illness where symptom relief, communication and goal-concordant care are central at any stage.

Pathophysiology. Disease and treatment effects combine with uncertainty, family/social roles and existential concerns.

Determinants and modifiers. Severity, prognosis, access, communication, culture and caregiver capacity.

Assessment. Symptoms, function, understanding, goals, caregiver burden and anticipatory needs.

Intervention logic. Concurrent palliative care, symptom treatment, advance care planning and hospice where appropriate.

Known limits and open questions. Prognosis is uncertain and palliative care is wrongly equated with abandonment.

Suite links. 14 interventions; 06 existential; 15 ethics.

Evidence anchors: ^[3,25,38]

5.0.13 Flourishing and positive health

Definition and phenotype. Capacity to pursue meaningful goals, relationships and participation with adaptive physical and mental functioning.

Pathophysiology. Biology, security, opportunity, skills, connection and purpose interact.

Determinants and modifiers. Resources, safety, social connection, autonomy, health behaviors and environment.

Assessment. Positive well-being, capability, participation and self-defined goals alongside conventional health measures.

Intervention logic. Upstream conditions, prevention, connection and opportunities rather than optimization products or compulsory positivity.

Known limits and open questions. Well-being measures are culturally sensitive and can obscure structural deprivation.

Suite links. 06 psychosocial; 11 public health; 15 equity.

Evidence anchors: ^[3,19,26]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current whole-person health states and outcomes risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

Continued on next page

Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current whole-person health states and outcomes risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[16]

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[4,5]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[16]

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[5,6]

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[31,33]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Narrative history and longitudinal timeline	Noninvasive	Person/caregiver account linked to records, exposures, treatments and events.	Meaning, chronology, hypotheses and priorities.	Integrates fragmented data and captures lived experience.	Recall, language, power and documentation bias.
Physical and mental-status examination	Noninvasive	Observed signs, performance, interaction and focused maneuvers.	Current phenotype, urgency and test selection.	Low-cost whole-person context.	Observer variation, masking/compensation and setting effects.
Patient-reported outcome measures	Noninvasive	Validated self-report scales for symptoms, function, well-being and participation.	Severity, interference, change and treatment value.	Direct access to experiences unavailable to devices.	Language/culture, response shift, burden and score abstraction.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
WHODAS/ICF-based disability assessment	Noninvasive	Structured self/interviewer report across cognition, mobility, self-care, relationships, life activities and participation.	Activity/participation limitation and support needs.	Cross-condition functional comparison.	Context, assistance and environment can be under-represented if used mechanically.
Performance-based capacity tests	Noninvasive	Timed walk, strength, balance, cognition or task completion under standardized conditions.	Capacity, reserve and change.	Objective task-level information.	Learning, motivation, pain, assistive devices and lab-real-world gap.
Ecological momentary assessment	Noninvasive digital/self-report	Brief in-context symptom, mood, exposure and activity sampling.	Within-person variability, triggers and temporal order.	Reduces long recall and captures episodes.	Prompt reactivity, missingness, burden and phone access.
Wearable and ambient monitoring	Noninvasive	Motion, sleep, cardiovascular, temperature and environmental sensors.	Activity, routine, physiology and change detection.	Longitudinal real-world context.	Algorithm drift, nonwear, privacy and disability/skin bias.
Laboratory and imaging summary profile	Minimally invasive/noninvasive	Organ-specific assays and imaging selected hypotheses.	Biologic state and disease evidence.	Mechanistic and prognostic information.	Incidental findings, overtesting and surrogate dominance.
Medication and intervention burden inventory	Noninvasive, informatics	Reconciled drugs, devices, procedures, self-care time, cost and side effects.	Treatment workload, interactions and adherence context.	Makes iatrogenic exposure visible.	Incomplete records and self-medication.
Social and environmental context assessment	Noninvasive	Housing, food, work, transport, safety, connection, discrimination and exposure measures.	Needs, resources and causal constraints.	Identifies upstream intervention targets.	Privacy, stigma, lack of referral resources and individualization of structural problems.
Goal-attainment and preference elicitation	Noninvasive	Structured conversation, decision aids and individualized outcome scaling.	Priorities, tradeoffs and goal-concordance.	Makes value component explicit.	Framing, health literacy, unstable preferences and proxy conflict.
Composite risk and frailty models	Software/noninvasive	Weighted clinical, lab, functional and social variables.	Probability of defined outcome or vulnerability.	Supports triage and communication.	Opaque weights, dataset shift, circularity and denial of care.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Integrated health record / personal health graph	Software	Time-linked diagnoses, raw/derived measurements, exposures, treatments, goals and provenance.	Cross-domain timeline and computable profile.	Can expose interactions and support continuity.	Semantic mismatch, data overload, privacy and incorrect inference.

7.0.1 Narrative history and longitudinal timeline

Measured quantity and transducer. Person/caregiver account linked to records, exposures, treatments and events.

Typical outputs. Meaning, chronology, hypotheses and priorities.

Strengths. Integrates fragmented data and captures lived experience.

Limitations and artifacts. Recall, language, power and documentation bias.

Reference and validation strategy. Structured chronology, corroboration where needed and repeated revision.

Clinical/research maturity. Established core.

Evidence anchors: [4,24]

7.0.2 Physical and mental-status examination

Measured quantity and transducer. Observed signs, performance, interaction and focused maneuvers.

Typical outputs. Current phenotype, urgency and test selection.

Strengths. Low-cost whole-person context.

Limitations and artifacts. Observer variation, masking/compensation and setting effects.

Reference and validation strategy. Standardization, repeated measures and outcome correlation.

Clinical/research maturity. Established.

Evidence anchors: [5,27]

7.0.3 Patient-reported outcome measures

Measured quantity and transducer. Validated self-report scales for symptoms, function, well-being and participation.

Typical outputs. Severity, interference, change and treatment value.

Strengths. Direct access to experiences unavailable to devices.

Limitations and artifacts. Language/culture, response shift, burden and score abstraction.

Reference and validation strategy. Content validity, measurement invariance, responsiveness and meaningful-change anchors.

Clinical/research maturity. Established.

Evidence anchors: ^[3,4]

7.0.4 WHODAS/ICF-based disability assessment

Measured quantity and transducer. Structured self/interviewer report across cognition, mobility, self-care, relationships, life activities and participation.

Typical outputs. Activity/participation limitation and support needs.

Strengths. Cross-condition functional comparison.

Limitations and artifacts. Context, assistance and environment can be underrepresented if used mechanically.

Reference and validation strategy. Validated administration and linkage to goals/environment.

Clinical/research maturity. Established.

Evidence anchors: ^[5,6]

7.0.5 Performance-based capacity tests

Measured quantity and transducer. Timed walk, strength, balance, cognition or task completion under standardized conditions.

Typical outputs. Capacity, reserve and change.

Strengths. Objective task-level information.

Limitations and artifacts. Learning, motivation, pain, assistive devices and lab-real-world gap.

Reference and validation strategy. Standard protocol, reliability and participation/outcome validation.

Clinical/research maturity. Established.

Evidence anchors: [4,9]

7.0.6 Ecological momentary assessment

Measured quantity and transducer. Brief in-context symptom, mood, exposure and activity sampling.

Typical outputs. Within-person variability, triggers and temporal order.

Strengths. Reduces long recall and captures episodes.

Limitations and artifacts. Prompt reactivity, missingness, burden and phone access.

Reference and validation strategy. Sampling/adherence analysis and external sensor/event validation.

Clinical/research maturity. Research/emerging clinical.

Evidence anchors: [30,31]

7.0.7 Wearable and ambient monitoring

Measured quantity and transducer. Motion, sleep, cardiovascular, temperature and environmental sensors.

Typical outputs. Activity, routine, physiology and change detection.

Strengths. Longitudinal real-world context.

Limitations and artifacts. Algorithm drift, nonwear, privacy and disability/skin bias.

Reference and validation strategy. Raw data/quality, reference tasks, subgroup and utility validation.

Clinical/research maturity. Emerging.

Evidence anchors: [17,30]

7.0.8 Laboratory and imaging summary profile

Measured quantity and transducer. Organ-specific assays and imaging selected by hypotheses.

Typical outputs. Biologic state and disease evidence.

Strengths. Mechanistic and prognostic information.

Limitations and artifacts. Incidental findings, overtesting and surrogate dominance.

Reference and validation strategy. Pretest-driven pathways and patient-important outcome links.

Clinical/research maturity. Established tools; integration challenging.

Evidence anchors: [16,24]

7.0.9 Medication and intervention burden inventory

Measured quantity and transducer. Reconciled drugs, devices, procedures, self-care time, cost and side effects.

Typical outputs. Treatment workload, interactions and adherence context.

Strengths. Makes iatrogenic exposure visible.

Limitations and artifacts. Incomplete records and self-medication.

Reference and validation strategy. Source reconciliation, patient confirmation and adverse-event follow-up.

Clinical/research maturity. Established but underused.

Evidence anchors: [24,32,33]

7.0.10 Social and environmental context assessment

Measured quantity and transducer. Housing, food, work, transport, safety, connection, discrimination and exposure measures.

Typical outputs. Needs, resources and causal constraints.

Strengths. Identifies upstream intervention targets.

Limitations and artifacts. Privacy, stigma, lack of referral resources and individualization of structural problems.

Reference and validation strategy. Consent, validated measures and closed-loop resource/action evaluation.

Clinical/research maturity. Established/emerging.

Evidence anchors: [19,20]

7.0.11 Goal-attainment and preference elicitation

Measured quantity and transducer. Structured conversation, decision aids and individualized outcome scaling.

Typical outputs. Priorities, tradeoffs and goal-concordance.

Strengths. Makes value component explicit.

Limitations and artifacts. Framing, health literacy, unstable preferences and proxy conflict.

Reference and validation strategy. Decision-quality, comprehension and longitudinal concordance.

Clinical/research maturity. Established selected.

Evidence anchors: ^[25,37]

7.0.12 Composite risk and frailty models

Measured quantity and transducer. Weighted clinical, lab, functional and social variables.

Typical outputs. Probability of defined outcome or vulnerability.

Strengths. Supports triage and communication.

Limitations and artifacts. Opaque weights, dataset shift, circularity and denial of care.

Reference and validation strategy. External calibration, subgroup/decision analysis and human review.

Clinical/research maturity. Established/emerging.

Evidence anchors: ^[17]

7.0.13 Integrated health record / personal health graph

Measured quantity and transducer. Time-linked diagnoses, raw/derived measurements, exposures, treatments, goals and provenance.

Typical outputs. Cross-domain timeline and computable profile.

Strengths. Can expose interactions and support continuity.

Limitations and artifacts. Semantic mismatch, data overload, privacy and incorrect inference.

Reference and validation strategy. FHIR/terminology mapping, provenance, user correction and workflow utility.

Clinical/research maturity. Infrastructure/emerging.

Evidence anchors: ^[33,34,35,36]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise whole-person health states and outcomes quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Continued on next page

Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise whole-person health states and outcomes quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: ^[30]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[31,33]

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[31]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[17]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[17]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[37]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Upstream prevention and healthy environments	Population/prevention	Reduce causal exposures and create feasible opportunities for movement, sleep, nutrition, safety and connection.	Broad evidence for major risk reduction when interventions are structural and sustained.	Blame, unequal access and unintended displacement.	Exposure, behavior opportunity, incidence, function and equity.
Shared decision and goal-oriented care	Clinical process	Align actions with goals, absolute benefit, burden and uncertainty.	Improves decision quality and can reduce unwanted care.	Time, low health literacy and performative choice without real options.	Comprehension, concordance, regret, outcomes and burden.
Integrated multimorbidity management	Clinical/system	Coordinate conditions, medicines, appointments and monitoring around total benefit.	Rational where single-disease plans conflict; evidence for models is context-dependent.	Care fragmentation and role ambiguity.	Function, adverse events, treatment workload, utilization and goal attainment.
Rehabilitation and adaptive support	Rehabilitation	Restore or compensate function through training, assistive products and environment.	Improves activity and participation across diagnoses.	Access, abandonment and inappropriate intensity.	Capability, participation, use and person-defined outcomes.
Symptom and pain management	Treatment/support	Address causal and symptomatic pathways while preserving function and safety.	Effective options exist but benefit is mechanism- and person-specific.	Polypharmacy, dependence, sedation and dismissal of causes.	Symptoms, interference, function, adverse effects and goals.
Mental-health and behavioral intervention	Treatment/with person	Reduce mood, anxiety, trauma, cognition and behavior as integrated health components.	Evidence-based therapies improve mental and physical outcomes in selected contexts.	Stigma, fragmented access and overpsychologizing.	Symptoms, safety, function, physical outcomes and participation.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Social and material support	Upstream/clinical system	Connect to food, housing, transport, legal, income, caregiving and violence resources while advocating structural change.	Can improve access and selected health outcomes when resources exist.	Screening without services and privacy harm.	Resource receipt, stability, health/function and equity.
Medication review and deprescribing	Safety/therapeutic	Remove nonbeneficial or harmful drugs while preserving essential treatment.	Reduces selected adverse effects and burden in appropriate patients.	Withdrawal, relapse and fragmented prescriber responsibility.	Symptoms, disease control, adverse events, function and regimen burden.
Caregiver and relationship support	Social/system	Support informal caregivers, communication, respite and shared coping.	Important to adherence, function and home care.	Burnout, coercion and unequal gendered burden.	Caregiver strain, safety, patient outcomes and relationship quality.
Palliative and serious-illness care	Supportive	Relieve symptoms, communicate uncertainty and align treatment with goals throughout serious illness.	Improves symptom and care experience; some settings show utilization/survival benefits.	Late referral and inequitable access.	Symptoms, goal concordance, caregiver outcomes and unwanted interventions.
Learning health-system follow-up	System	Use structured outcome feedback and incident learning to adapt care.	Can improve quality when data become action rather than surveillance.	Measurement burden, gaming and inequitable algorithmic control.	Outcome improvement, safety, burden and subgroup effects.

9.0.1 Upstream prevention and healthy environments

Target and mechanism. Reduce causal exposures and create feasible opportunities for movement, sleep, nutrition, safety and connection.

Evidence and expected outcome. Broad evidence for major risk reduction when interventions are structural and sustained.

Risks, burden, and equity. Blame, unequal access and unintended displacement.

Response monitoring. Exposure, behavior opportunity, incidence, function and equity.

Evidence anchors: [19,20,39]

9.0.2 Shared decision and goal-oriented care

Target and mechanism. Align actions with goals, absolute benefit, burden and uncertainty.

Evidence and expected outcome. Improves decision quality and can reduce unwanted care.

Risks, burden, and equity. Time, low health literacy and performative choice without real options.

Response monitoring. Comprehension, concordance, regret, outcomes and burden.

Evidence anchors: [25,37]

9.0.3 Integrated multimorbidity management

Target and mechanism. Coordinate conditions, medicines, appointments and monitoring around total benefit.

Evidence and expected outcome. Rational where single-disease plans conflict; evidence for models is context-dependent.

Risks, burden, and equity. Care fragmentation and role ambiguity.

Response monitoring. Function, adverse events, treatment workload, utilization and goal attainment.

Evidence anchors: [22,24]

9.0.4 Rehabilitation and adaptive support

Target and mechanism. Restore or compensate function through training, assistive products and environment.

Evidence and expected outcome. Improves activity and participation across diagnoses.

Risks, burden, and equity. Access, abandonment and inappropriate intensity.

Response monitoring. Capability, participation, use and person-defined outcomes.

Evidence anchors: [5,9,10]

9.0.5 Symptom and pain management

Target and mechanism. Address causal and symptomatic pathways while preserving function and safety.

Evidence and expected outcome. Effective options exist but benefit is mechanism- and person-specific.

Risks, burden, and equity. Polypharmacy, dependence, sedation and dismissal of causes.

Response monitoring. Symptoms, interference, function, adverse effects and goals.

Evidence anchors: [4,29]

9.0.6 Mental-health and behavioral intervention

Target and mechanism. Treat mood, anxiety, trauma, cognition and behavior as integrated health components.

Evidence and expected outcome. Evidence-based therapies improve mental and physical outcomes in selected contexts.

Risks, burden, and equity. Stigma, fragmented access and overpsychologizing.

Response monitoring. Symptoms, safety, function, physical outcomes and participation.

Evidence anchors: [27,28]

9.0.7 Social and material support

Target and mechanism. Connect to food, housing, transport, legal, income, caregiving and violence resources while advocating structural change.

Evidence and expected outcome. Can improve access and selected health outcomes when resources exist.

Risks, burden, and equity. Screening without services and privacy harm.

Response monitoring. Resource receipt, stability, health/function and equity.

Evidence anchors: [19,23]

9.0.8 Medication review and deprescribing

Target and mechanism. Remove nonbeneficial or harmful drugs while preserving essential treatment.

Evidence and expected outcome. Reduces selected adverse effects and burden in appropriate patients.

Risks, burden, and equity. Withdrawal, relapse and fragmented prescriber responsibility.

Response monitoring. Symptoms, disease control, adverse events, function and regimen burden.

Evidence anchors: [24]

9.0.9 Caregiver and relationship support

Target and mechanism. Support informal caregivers, communication, respite and shared coping.

Evidence and expected outcome. Important to adherence, function and home care.

Risks, burden, and equity. Burnout, coercion and unequal gendered burden.

Response monitoring. Caregiver strain, safety, patient outcomes and relationship quality.

Evidence anchors: [3,26]

9.0.10 Palliative and serious-illness care

Target and mechanism. Relieve symptoms, communicate uncertainty and align treatment with goals throughout serious illness.

Evidence and expected outcome. Improves symptom and care experience; some settings show utilization/survival benefits.

Risks, burden, and equity. Late referral and inequitable access.

Response monitoring. Symptoms, goal concordance, caregiver outcomes and unwanted interventions.

Evidence anchors: [3,25]

9.0.11 Learning health-system follow-up

Target and mechanism. Use structured outcome feedback and incident learning to adapt care.

Evidence and expected outcome. Can improve quality when data become action rather than surveillance.

Risks, burden, and equity. Measurement burden, gaming and inequitable algorithmic control.

Response monitoring. Outcome improvement, safety, burden and subgroup effects.

Evidence anchors: [22,40]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can whole-person health be summarized without losing meaning?	Domain scores and composites support comparison and triage.	A transportable score that preserves causal differences, preferences and equity.	Compensation among dimensions can hide severe problems and weights are value choices.	Report a profile/dashboard with explicit goals and test whether any composite improves decisions.
How should response shift be interpreted?	People adapt and may change standards or values over time.	When improved self-report reflects true recovery, adaptation or constrained expectation.	External function and internal meaning can legitimately diverge.	Mixed qualitative, individualized goal and objective longitudinal measures.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within whole-person health states and outcomes.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Continued on next page

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can whole-person health be summarized without losing meaning?

Established knowledge. Domain scores and composites support comparison and triage.

Unresolved issue. A transportable score that preserves causal differences, preferences and equity.

Why current evidence is insufficient. Compensation among dimensions can hide severe problems and weights are value choices.

Research that would change confidence. Report a profile/dashboard with explicit goals and test whether any composite improves decisions.

Evidence anchors: ^[3,17]

10.0.2 How should response shift be interpreted?

Established knowledge. People adapt and may change standards or values over time.

Unresolved issue. When improved self-report reflects true recovery, adaptation or constrained expectation.

Why current evidence is insufficient. External function and internal meaning can legitimately diverge.

Research that would change confidence. Mixed qualitative, individualized goal and objective longitudinal measures.

Evidence anchors: ^[3,4]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within whole-person health states and outcomes.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[15]

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[4,17]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[16]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[30]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[4,5]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Whole-person common outcome set	Which minimal measures preserve mortality, symptoms, function, participation, burden and goals across diseases?	Participatory Delphi plus psychometric and pragmatic validation in diverse multimorbidity cohorts.	Low-burden set predicts decisions and remains invariant/meaningful.	Administrative convenience outweighs person value.
0. Goal-concordant multimorbidity trial	Does a health-graph and goal review improve total benefit versus guideline-by-guideline care?	Pragmatic cluster randomized trial with function, adverse events, burden, equity and cost.	Improved goal attainment/function with no safety loss.	Fewer metrics but no better lived outcome.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for whole-person health states and outcomes?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.

Continued on next page

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Whole-person common outcome set

Question. Which minimal measures preserve mortality, symptoms, function, participation, burden and goals across diseases?

Design. Participatory Delphi plus psychometric and pragmatic validation in diverse multimorbidity cohorts.

Success criterion. Low-burden set predicts decisions and remains invariant/meaningful.

Failure or stop criterion. Administrative convenience outweighs person value.

Evidence anchors: [4,5,37]

11.0.2 0. Goal-concordant multimorbidity trial

Question. Does a health-graph and goal review improve total benefit versus guideline-by-guideline care?

Design. Pragmatic cluster randomized trial with function, adverse events, burden, equity and cost.

Success criterion. Improved goal attainment/function with no safety loss.

Failure or stop criterion. Fewer metrics but no better lived outcome.

Evidence anchors: [22,40]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for whole-person health states and outcomes?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[21,31]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

Evidence anchors: ^[15,18]

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[17]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [16,22]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: [5,36]

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: [21,22]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
030	Integrated Homeostatic Systems	Reserve, challenge, stability and recovery connect organ physiology to whole-person trajectories.	resting measures do not explain vulnerability or exertion.
05	Brain, Cognition, Emotion and Behavior	Symptoms, cognition, mood, behavior and meaning are whole-person outcomes and mechanisms.	mental and physical domains interact.
06	Psychosocial, Relational, Cultural and Existential Health	Relationships, culture, safety, identity and purpose shape health experience.	context or meaning changes the phenotype.
10	Function, Disability, Rehabilitation and Participation	Provides the detailed capability, environment and rehabilitation model.	impairment must be translated into activity and participation.
12	Health Care and Delivery Systems	Care burden, access, coordination and safety become determinants and outcomes.	the care system itself affects health.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Capability	Real opportunity to achieve valued activities, influenced by body, resources and environment.
Participation	Involvement in life situations and social roles.
Treatment burden	Work, time, cost, adverse effects and cognitive/emotional load imposed by healthcare.
Response shift	Change in internal standards, values or conceptualization used to judge quality of life.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

- [1] Technical Evidence and Systems Review. Atlas of Human Health: A Unified Taxonomy. 2026. Master taxonomy and conceptual architecture for this suite.
- [2] Technical Evidence and Engineering Review. Cardiovascular Health and Measurement Engineering. 2026. Volume 03A of this suite.
- [3] World Health Organization. WHOQOL: Measuring Quality of Life. 2026. <https://www.who.int/tools/whoqol>
- [4] National Institutes of Health. Patient-Reported Outcomes Measurement Information System (PROMIS). 2026. <https://commonfund.nih.gov/promis>
- [5] World Health Organization. International Classification of Functioning, Disability and Health. 2001/2026. <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health>
- [6] World Health Organization. WHO Disability Assessment Schedule 2.0. 2012/2026. <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health/who-disability-assessment-schedule>
- [7] United Nations. Convention on the Rights of Persons with Disabilities. 2006. <https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities>
- [8] World Health Organization. Global report on health equity for persons with disabilities. 2022. <https://www.who.int/publications/i/item/9789240063600>
- [9] World Health Organization. Rehabilitation 2030 initiative. 2026. <https://www.who.int/initiatives/rehabilitation-2030>
- [10] World Health Organization and UNICEF. Global report on assistive technology. 2022/2026. <https://www.who.int/publications/i/item/9789240049451>
- [11] GBD 2023 Collaborators. Burden of diseases and injuries, risk-attributable burden, and healthy life expectancy, 1990-2023. 2025. <https://www.healthdata.org/research-analysis/gbd>
- [12] Lopez-Otin C, Blasco MA, Partridge L, Serrano M, Kroemer G. Hallmarks of aging: an expanding universe. 2023. <https://doi.org/10.1016/j.cell.2022.11.001>
- [13] McEwen BS, Stellar E. Stress and the individual: mechanisms leading to disease. 1993. <https://doi.org/10.1001/archinte.1993.00410180039004>
- [14] World Health Organization. Life course approach to health. 2025/2026. <https://www.who.int/our-work/life-course>
- [15] Hernan MA, Robins JM. Causal Inference: What If. 2020. <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>
- [16] Guyatt GH, Oxman AD, Vist GE, et al.. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. 2008. <https://doi.org/10.1136/bmj.39489.470347.AD>

- [17] Collins GS, Moons KGM, Dhiman P, et al.. TRIPOD+AI statement. 2024. <https://www.tripod-statement.org/>
- [18] Hill AB. The environment and disease: association or causation?. 1965. <https://www.jstor.org/stable/25491524>
- [19] World Health Organization. World report on social determinants of health equity. 2025. <https://www.who.int/publications/i/item/9789240107588>
- [20] World Health Organization. Health and Environment Country Scorecards, 2025 edition. 2025. <https://www.who.int/teams/environment-climate-change-and-health/monitoring/scorecards>
- [21] World Health Organization. Health equity. 2026. <https://www.who.int/health-topics/health-equity>
- [22] Damschroder LJ, Reardon CM, Widerquist MAO, Lowery J. The updated Consolidated Framework for Implementation Research. 2022. <https://doi.org/10.1186/s13012-022-01245-0>
- [23] World Health Organization and World Bank. Tracking Universal Health Coverage: 2025 Global Monitoring Report. 2025. <https://www.who.int/publications/i/item/9789240117808>
- [24] World Health Organization. Global Patient Safety Action Plan 2021-2030. 2021. <https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan>
- [25] World Health Organization. Palliative care. 2025. <https://www.who.int/news-room/fact-sheets/detail/palliative-care>
- [26] World Health Organization Commission on Social Connection. From loneliness to social connection: charting a path to healthier societies. 2025. <https://www.who.int/publications/i/item/9789240112681>
- [27] American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision. 2022/2025. <https://www.psychiatry.org/psychiatrists/practice/dsm>
- [28] National Institute of Mental Health. Research Domain Criteria (RDoC). 2026. <https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc>
- [29] International Association for the Study of Pain. IASP terminology and ICD-11 pain classification. 2025/2026. <https://www.iasp-pain.org/resources/terminology/>
- [30] U.S. Food and Drug Administration. Digital Health Center of Excellence and current guidance. 2026. <https://www.fda.gov/medical-devices/digital-health-center-excellence>
- [31] Wilkinson MD, Dumontier M, Aalbersberg IJ, et al.. The FAIR Guiding Principles for scientific data management and stewardship. 2016. <https://doi.org/10.1038/sdata.2016.18>
- [32] World Health Organization. International Classification of Health Interventions. 2025/2026. <https://www.who.int/standards/classifications/international-classification-of-health-interventions>
- [33] HL7 International. FHIR Release 5 specification. 2023/2026. <https://hl7.org/fhir/>
- [34] SNOMED International. SNOMED CT. 2026. <https://www.snomed.org/>
- [35] Regenstrief Institute. LOINC: standard identifiers for health measurements, observations, and documents. 2026. <https://loinc.org/>
- [36] OHDSI. OMOP Common Data Model. 2026. <https://www.ohdsi.org/data-standardization/>
- [37] National Commission for the Protection of Human Subjects. The Belmont Report. 1979. <https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/>
- [38] Office of the UN High Commissioner for Human Rights. International standards on the right to physical and mental health. 2026. <https://www.ohchr.org/en/special-procedures/sr-health/international-standards-right-physical-and-mental-health>

- [39] World Health Organization. Best buys and other recommended interventions for prevention and control of noncommunicable diseases. 2023. <https://www.who.int/publications/i/item/9789240091078>
- [40] Institute of Medicine. Best Care at Lower Cost: The Path to Continuously Learning Health Care in America. 2013. <https://doi.org/10.17226/13444>

Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.